

Kira College
CHEMISTRY 12 SEMESTER 1 EXAMINATION 2015
Booklet 2



Name: _____ Teacher: _____

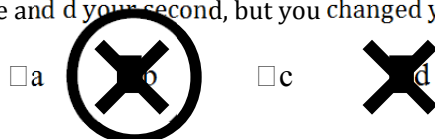
For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. For example, if b is your answer: ☐ a ☒ b ☐ c ☐ d

If you make a mistake, place a cross through that square, do not erase or use correction fluid. Shade your new answer.

For example, if b is a mistake and d is your correct answer: ☐ a ☒ b ☐ c ☒ d

In the event that you then change your mind back to your original answer, you then cross out the second selection and then circle the first choice.

For example, if b was the first choice and d your second, but you changed your mind back and b is your answer:



Multiple Choice: Mark the answer by shading in the box to the LEFT of the letter you chose.

- | | |
|---|---|
| 1. ☐ a ☐ b ☐ c ☐ d | 14. ☐ a ☐ b ☐ c ☐ d |
| 2. ☐ a ☐ b ☐ c ☐ d | 15. ☐ a ☐ b ☐ c ☐ d |
| 3. ☐ a ☐ b ☐ c ☐ d | 16. ☐ a ☐ b ☐ c ☐ d |
| 4. ☐ a ☐ b ☐ c ☐ d | 17. ☐ a ☐ b ☐ c ☐ d |
| 5. ☐ a ☐ b ☐ c ☐ d | 18. ☐ a ☐ b ☐ c ☐ d |
| 6. ☐ a ☐ b ☐ c ☐ d | 19. ☐ a ☐ b ☐ c ☐ d |
| 7. ☐ a ☐ b ☐ c ☐ d | 20. ☐ a ☐ b ☐ c ☐ d |
| 8. ☐ a ☐ b ☐ c ☐ d | 21. ☐ a ☐ b ☐ c ☐ d |
| 9. ☐ a ☐ b ☐ c ☐ d | 22. ☐ a ☐ b ☐ c ☐ d |
| 10. ☐ a ☐ b ☐ c ☐ d | 23. ☐ a ☐ b ☐ c ☐ d |
| 11. ☐ a ☐ b ☐ c ☐ d | 24. ☐ a ☐ b ☐ c ☐ d |
| 12. ☐ a ☐ b ☐ c ☐ d | 25. ☐ a ☐ b ☐ c ☐ d |
| 13. ☐ a ☐ b ☐ c ☐ d | |

Section Two: Short answer**35% (70 marks)**

This section has **10** questions. Answer **all** questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three (3) significant figures and include appropriate units where applicable.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

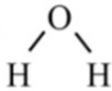
Suggested working time: 70 minutes

Question 26**(9 marks)**

Complete the table below by drawing the Lewis structures of the molecules listed and either drawing or naming the shape of the molecule.

All valence shell electron pairs should be represented either as : or as — .

The first row has been completed as an example.

Molecule	Lewis diagram	Sketch or name of shape	Number of <u>unbonded</u> electrons in molecule
H ₂ O	$\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$	bent, or 	4 (or 2 pairs)
PH ₃			
CF ₄			
SO ₃			

See next page

(4 marks)

[illegible]

(10 marks)

(a) For each of these molecules, choose the description that best describes the polarity of the covalent bond and molecule. (2 marks)

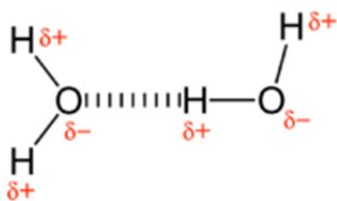
Molecule	Polarity of covalent bonds (write polar or non-polar)	Polarity of molecule (write polar or non-polar)
NH ₃		
SO ₃		

(b) For each of the following solids, state the **most dominant** form of intermolecular attraction and rank them in **increasing** order of melting point (i.e. 1 for lowest melting point and 4 for highest melting point). (4 marks)

Solid	Name of most dominant intermolecular force in solid state	Relative melting point ranking (write 1, 2, 3 or 4)
NH ₃		
PH ₃		
AsH ₃		
H ₂ O		

See next page

(c) The diagram below shows the physical attraction between neighbouring water molecules.



Sodium fluoride (NaF) and hydrogen fluoride are substances that are soluble in water.

i. Use the diagram above as a guide to show the **most dominant** form of physical attraction that exist between

- sodium fluoride and water molecules
- hydrogen fluoride and water molecules.

Your diagrams should include dotted lines (----) to show the relevant attraction as well as the location of charge densities (δ^+ and δ^-)

ii. Name the type of physical interaction in each case e.g. dispersion forces. (4 marks)

Sodium fluoride and water molecules	Hydrogen fluoride and water molecules
Name of most dominant attractive force	Name of most dominant attractive force

Question 29

(4 marks)

For the following reactions, describe fully the **observed changes**, including any:

- colour changes
- odours
- precipitates (give the colour)
- gases evolved (give the colour or describe as colourless).

(a) A sealed tube containing the equilibrium mixture, $2\text{NO}_{2(g)} \leftrightarrow \text{N}_2\text{O}_{4(g)}$ $\Delta H = -90 \text{ kJmol}^{-1}$ is placed in a beaker of hot water. (1 mark)

(b) Barium nitrate solution is added to potassium phosphate solution. (1 mark)

(c) Copper(II) chloride solid is added to sodium carbonate solution. (2 marks)

See next page

Question 30**(4 marks)**

Write balanced **ionic** equations to represent the reactions described below.

(a) Sodium hydroxide solution is added to iron(III) iodide solution.

(2 marks)

(b) Dilute hydrochloric acid is added to potassium chromate (K_2CrO_4) solution.

(2 marks)

Question 31**(6 marks)**

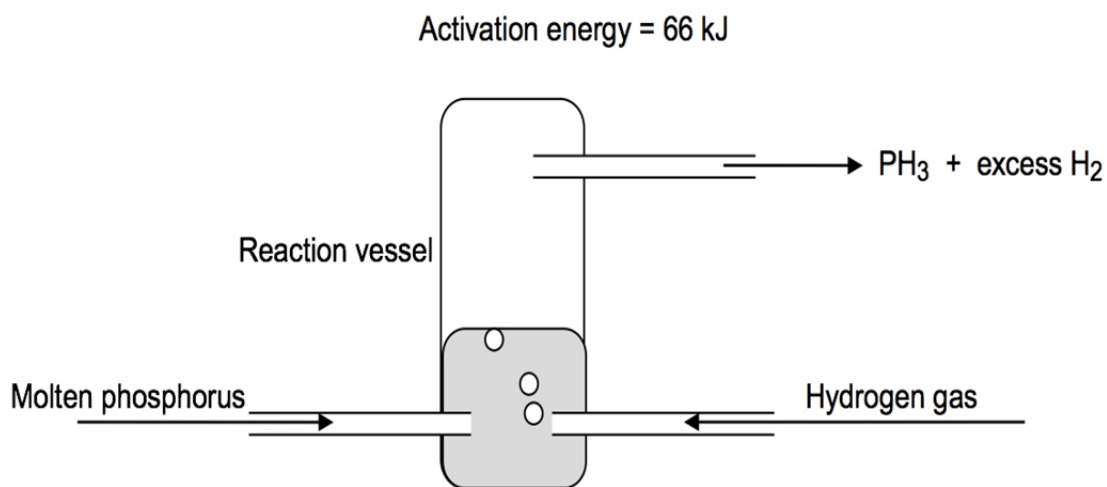
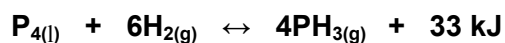
For each of the following pairs of substances, provide details of a **chemical** test that would allow the two substances to be distinguished from one another. Equations are **not** required.

Substances	Chemical test	Observations
$\text{Na}_2\text{SO}_{4(s)}$ and $\text{NaCH}_3\text{COO}_{(s)}$		with $\text{Na}_2\text{SO}_{4(s)}$: with $\text{NaCH}_3\text{COO}_{(s)}$:
$\text{KOH}_{(aq)}$ and $\text{K}_3\text{PO}_{4(aq)}$		with $\text{KOH}_{(aq)}$: with $\text{K}_3\text{PO}_{4(aq)}$:

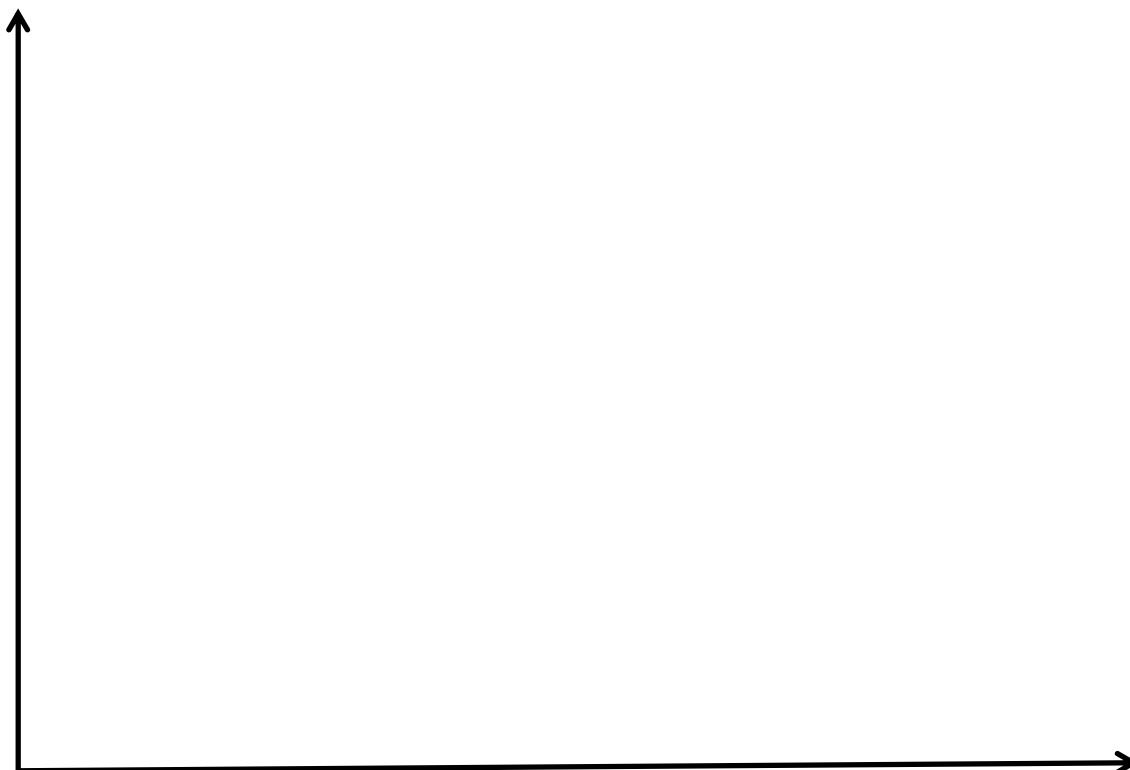
See next page

Question 32**(10 marks)**

Phosphine (PH_3) is a gas that can be produced by bubbling hydrogen gas through molten phosphorus. The reaction is reversible



- (a) Use the axes below to draw a labelled energy profile diagram to represent the process. State clearly the axes' headings, reactants, products, activation energy and change in enthalpy (ΔH). (4 marks)



- (b) Would a high temperature, or a low temperature, **most likely** be used in this industrial process?
(Circle one response below). (3 marks)

Moderate Temperature

Low Temperature

State the effect of temperature on the forward reaction rate and yield of this industrial process.

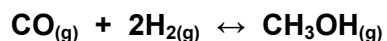
	High Temperature	Low Temperature
Effect on forward reaction rate (write increase, decrease or no change)		
Effect on yield (write increase, decrease or no change)		

- (c) Mixtures of ammonia and phosphine can be separated by distillation due to their different boiling points. Explain why the boiling point of phosphine is lower than that of ammonia even though its molar mass is nearly twice that of ammonia. (3 marks)

	Phosphine (PH ₃)	Ammonia (NH ₃)
Boiling point (°C)	- 88	- 33
Molar mass (gmol ⁻¹)	33.99	17.03

Question 33**(8 marks)**

Consider the equilibrium reaction below



$$\Delta H = -90 \text{ kJmol}^{-1}$$

Indicate in the table below whether there would be an increase, decrease, or no change in the concentration of $\text{CO}_{(\text{g})}$ after the changes given in the table are imposed on the system and **equilibrium has been re-established**. Provide a brief explanation for the observation using relevant knowledge of reaction rates and Collision Theory.

Change	Change in concentration of $\text{CO}_{(\text{g})}$ (circle the correct response)	Brief explanation
The volume of the reaction vessel is halved	. increase . decrease . no change	
The temperature of the reaction system is doubled	. increase . decrease . no change	
Neon gas (Ne) is pumped into the system while keeping the volume constant	. increase . decrease . no change	
H_2 gas is pumped into the system while keeping the volume constant	. increase . decrease . no change	

See next page

Question 34**(5 marks)**

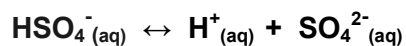
A number of ions can act as an acid or a base in certain conditions. These ions are said to be amphiprotic in nature.

- (a) Hydrogenphosphate (HPO_4^{2-}) is a polyatomic ion that can act as an acid or a base. Use a suitable chemical equation to show that HPO_4^{2-} can act as (2 marks)

i. a **Lowry-Bronsted acid**

ii. a **Lowry-Bronsted base**

- (b) The equilibrium constant (K) for the forward reaction of the equilibrium system below is 1.0×10^{-2} at room temperature.



Is HSO_4^- a strong or weak acid at room temperature? Circle one response below. (1 mark)

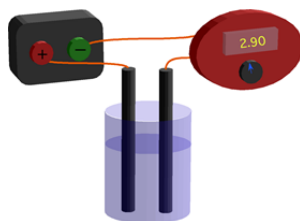
Strong acid

Weak acid

Explain your answer using relevant knowledge of acids and bases theories and equilibrium constant. (2 marks)

Question 35**(10 marks)**

In a Science experiment, a student measures the conductivity (in milliAmps) of three solutions of ethanoic acid. The diagram below shows her experimental set up.



The students' results are listed in the table below

Solution	pH	Electrical current reading (milliAmps)
1	2	20
2	3	9
3	4	4

- (a) **State** the relationship between pH and electrical conducting ability of ethanoic acid for this experiment. (1 mark)

- (b) **Explain** the relationship stated in part (a) using relevant knowledge of solutions and theories of acids and bases. (2 marks)

- (c) Would you expect a 0.1 mol L^{-1} solution of hydrochloric acid to have the same conductivity as that of a 0.1 mol L^{-1} solution of ethanoic acid? Explain your answer using relevant knowledge of the theories of acids and bases. (3 marks)

- (d) Phosphoric acid (H_3PO_4) is a weak triprotic acid.

List all the ions present (in order of decreasing concentration) in a 0.1 mol L^{-1} solution of phosphoric acid. **Explanation is not required.** (2 marks)

- (e) 900.0 mL of water is added to a 100.0 mL solution of a strong acid solution that has a pH of 6. Calculate the pH of the new solution. (2 marks)

Section Three: Extended Answers

40% (80 marks)

This section contains **four (4)** questions. You must answer **all** questions. Write your answers in the spaces provided.

Where questions require an explanation and/or description, marks are awarded for the relevant chemical content and also for coherence and clarity of expression. Lists or dot points are unlikely to gain full marks.

Final answers to calculations should be expressed to **three** significant figures.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

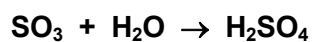
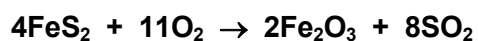
- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

Suggested working time: 80 minutes.

This space intentionally left blank

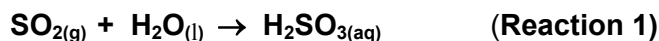
Question 36**(10 marks)**

The preparation of sulfuric acid (H_2SO_4) from iron pyrites (FeS_2) can be represented by the following series of reactions:

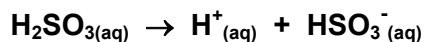


- (a) How many moles of sulfuric acid would be produced from **ONE** mole of iron pyrites? (1 mark)
- (b) What mass of iron pyrites would be needed to produce 123 tonne ($1 \text{ tonne} = 1 \times 10^6 \text{ g}$) of sulfuric acid if the Contact process only converts 95.0% of the sulfur dioxide and the other reactions achieve 100% conversion? (6 marks)

Sulfurous acid (H_2SO_3) can be produced by the dissolving of sulfur dioxide gas in liquid water under the right conditions.



H_2SO_3 is a weak acid that partially ionises to form hydrogen and hydrogen sulfite ions.



- (c) If **Reaction 1** above is slightly **exothermic** then predict the effect of the following changes on the pH of the solution (3 marks)

Change to equilibrium	Effect on pH (write increase, decrease or no change)
A little concentrated nitric acid is added	
Finely powdered $\text{NaOH}_{(s)}$ is added	
The temperature of the system is increased to a new constant value	

Question 37

(20 marks)

Magnesium, a reactive metal found in many common minerals, is also an essential nutrient for both plants and animals.

- (a) Write the electron configuration for magnesium ion (Mg^{2+}). (1 mark)

- (b) Draw the Lewis electron dot diagram for **MgCl_2**

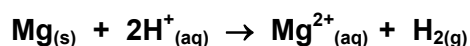
(1 mark)

- (c) Successive ionization energies of magnesium are given in the table below

	First	Second	Third
Energy required / kJmol^{-1}	738	1450	7730

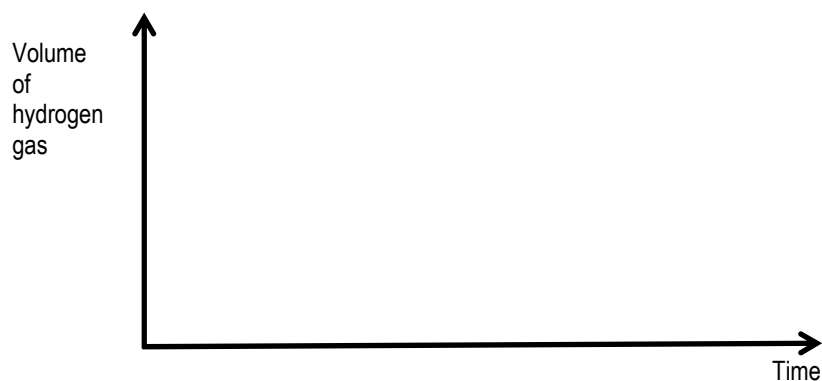
Explain why the third ionization energy is **much** greater than the second ionisation energy. (3 marks)

- (d) Magnesium metal is often used in school laboratories to make hydrogen gas. The equation for the reaction between dilute sulfuric acid and magnesium metal is



In a reaction rate experiment, a student studies the rate of the reaction above by measuring the volume of hydrogen gas produced over time. Assuming no significant errors are made, **sketch** a graph to show what the student is most likely to observe from the volume readings over time.

(1 mark)



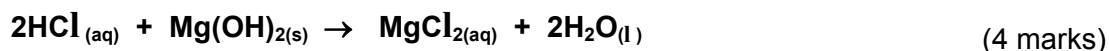
- (e) A well known compound of magnesium is magnesium hydroxide $\text{Mg}(\text{OH})_2$ which is commonly used as an antacid to neutralise stomach acid.

A typical magnesium hydroxide antacid tablet contains 300mg of magnesium hydroxide.

In the stomach, the normal concentration of hydrochloric acid is 0.100 mol L^{-1} . The volume of acid in the stomach can be assumed to be 0.650 litres. Calculate the number of moles of hydrochloric acid in the stomach. (1 mark)

A stressed-out student, studying for her Chemistry exam eats a whole packet of Tim Tams, causing the concentration of the acid in her stomach to rise to 0.200 mol L^{-1} . Calculate the number of moles of hydrochloric acid at this new concentration. (1 mark)

Calculate the number of magnesium hydroxide tablets the student would need to reduce the concentration of acid in her stomach back to the normal level. The non-ionic chemical reaction between stomach acid and magnesium hydroxide is:



- (g) Magnesium hydroxide is only sparingly soluble in water and the equilibrium below exists in a saturated solution.



Use relevant knowledge of reaction rates and Collision Theory to state and explain how changes in pH affect the **solubility** of solid magnesium hydroxide (i.e. more soluble or less soluble) in the equilibrium system above if the temperature is kept constant. (8 marks)

Change: Increase in pH		
Solubility of $\text{Mg}(\text{OH})_2$: (circle one response)	more soluble	less soluble
Explanation: _____		

Change: Decrease in pH		
Solubility of $\text{Mg}(\text{OH})_2$: (circle one response)	more soluble	less soluble
Explanation: _____		

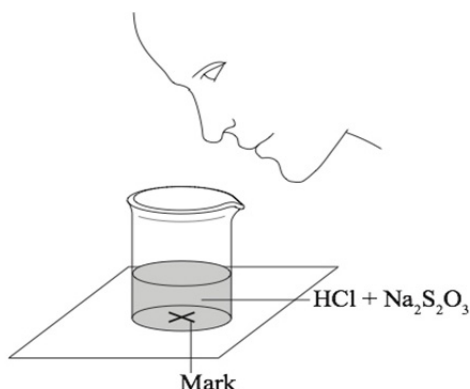
Question 38

(25 marks)

A group of students investigated the rate of reaction between aqueous sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) and hydrochloric acid according to the non-ionic chemical equation below.



The two reagents were rapidly mixed and placed over a mark on a piece of paper. The time taken for the precipitate of sulfur to obscure the mark 'X' (i.e. 'X' can no longer be seen) when viewed through the reaction mixture was recorded.



Initially the students measured out 10.0 mL of 0.500 mol L^{-1} hydrochloric acid and to this solution added 40.0 mL of $0.0200 \text{ mol L}^{-1}$ aqueous sodium thiosulfate. The mark on the paper was obscured 47 seconds after the solution was mixed.

- (a) The teacher asked the students to study the effect of **halving** the concentration of sodium thiosulfate on the rate of the reaction.

i. State the volumes of the liquids that should be mixed.

(2 marks)

Liquid	$0.500 \text{ mol L}^{-1} \text{ HCl}$	$0.0200 \text{ mol L}^{-1} \text{ Na}_2\text{S}_2\text{O}_3$	Water
Volume			

ii. State why it is important that the students use a similar beaker for both occasions.

(1 mark)

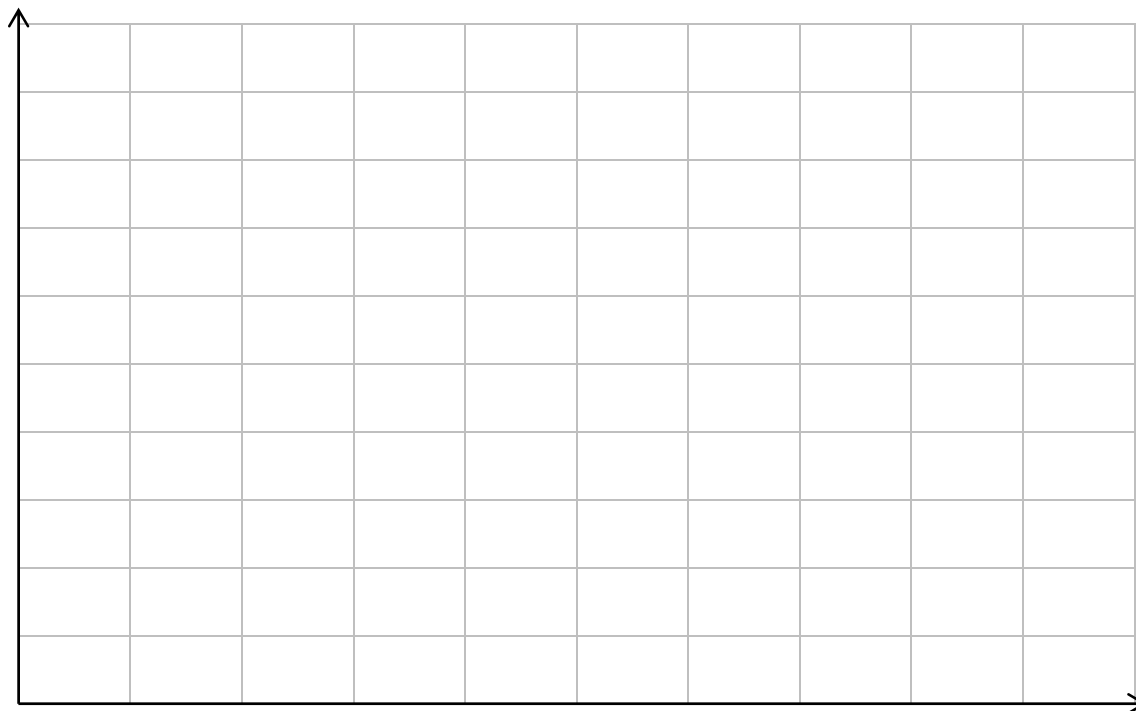
iii. State one (1) safety aspect that was neglected by the instructional diagram above.

(1 mark)

- iv. The results of a **separate** experiment investigating the effect of the concentration of hydrochloric acid of the reaction rate, while keeping the concentration of the sodium thiosulfate at the **original concentration value**, are given in the table below.

Concentration of HCl (molL ⁻¹)	0.020	0.040	0.060	0.080
Time (s)	89.1	72.8	62.4	54.2

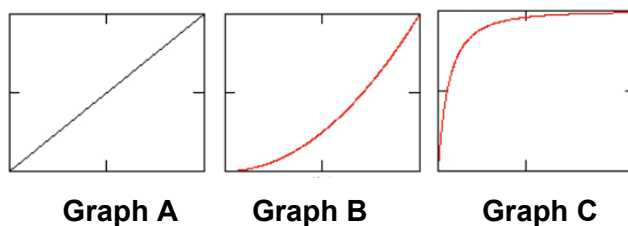
On the axes provided, draw a **fully labelled** graph to plot the information above. (4 marks)



- v. Explain the shape of the graph using relevant knowledge of Collision theory and reaction rates. (2 marks)

- (b) The teacher asked the students to devise another technique to measure the rate of this reaction. One group suggested recording how long it takes for the pH of the solution to change by one unit.

Which of the following graphs **best** represents the change in the pH of the reaction mixture as the experiment proceeds? (Circle one of the graphs below) (1 mark)



See next page

- (c) Another group of students suggested measuring the sulfur dioxide gas evolved over time.
- i. Calculate the volume of sulfur dioxide gas (in litres) that a mixture of 10.0 mL of 0.500 molL⁻¹ hydrochloric acid and 40.0 mL of 0.0200 molL⁻¹ aqueous sodium thiosulfate would produce at 101.3 kPa and 25°C. **Write your final numerical answer in 3 significant figures.** (5 marks)

- ii. Sulfur dioxide gas, a major cause of acid rain, is highly soluble in water. Use your knowledge of relevant bonding theories to explain why this is the case. You may find a suitable diagram useful in your explanation. (2 marks)

iii. The equilibrium equation for the dissolution of sulfur dioxide in water is shown below.



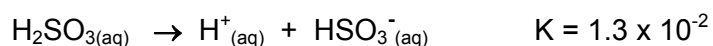
Write an equilibrium expression, K for the reaction above.

(1 mark)

K =

At 0°C, the solubility of sulfur dioxide in water is 228 gL⁻¹ but at a temperature of 90°C, its solubility in water is only 5.8 gL⁻¹. Is the dissolution of sulfur dioxide in water endothermic or exothermic? Explain your answer using relevant knowledge of Collision theory and reaction rates (4 marks)

iv. Sulfur dioxide can form two different acids in water, sulfurous acid (H₂SO₃) and sulfuric acid (H₂SO₄). The ionisation equation and the K value for each acid at 25°C is shown below.



Which of the two acids above is a better electrolyte at 25°C? **Explanation is not required.**

(1 mark)

Write a chemical equation to show that HSO₃⁻ can act as an **Arrhenius acid**.

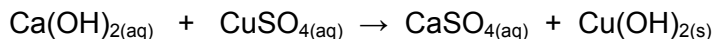
(1 mark)

Question 39**(25 marks)**

Mildews and fungi can cause great damage to grape vines. One chemical spray used to control these diseases is called Bordeaux mixture. A home gardener who wishes to treat his grapes with Bordeaux mixture prepares a mixture using the instructions given below.

- i. Add 25.0 g of calcium hydroxide powder to 25.0 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ powder.
- ii. Mix these powders with a small amount of water to make a paste.
- iii. Add the paste to 5.00 L of water and mix well.
- iv. Use the mixture immediately after preparation.

The non-ionic equation for the reaction that occurs after the powders are mixed with water is



- (a) Determine the limiting reactant for the above reaction. Show all working. (4 marks)

- (b) Calculate the **unused** mass of the excess reactant. (3 marks)

- (c) What colour (if any) will the precipitate have? _____ (1 mark)

Commercially, elemental sulfur (S) can be used to control outbreaks of powdery mildew on grapes.

- (d) Write the electron configuration for sulfide ion? _____ (1 mark)

- (e) Which of the following species would you expect to have a larger radius? (Circle **one** of the answers below).

A SULFUR ATOM

A SULFIDE ION

(1 mark)

See next page

(f) Explain your answer to part (e) using relevant knowledge of atomic structure and bonding. (3 marks)

Sulfur remaining on the grapes after harvest can be converted to a number of undesirable compounds during fermentation in wine production.

A wine chemist uses a method called Atomic Absorption Spectroscopy (AAS) to determine the amount of sulfur remaining on grapes.

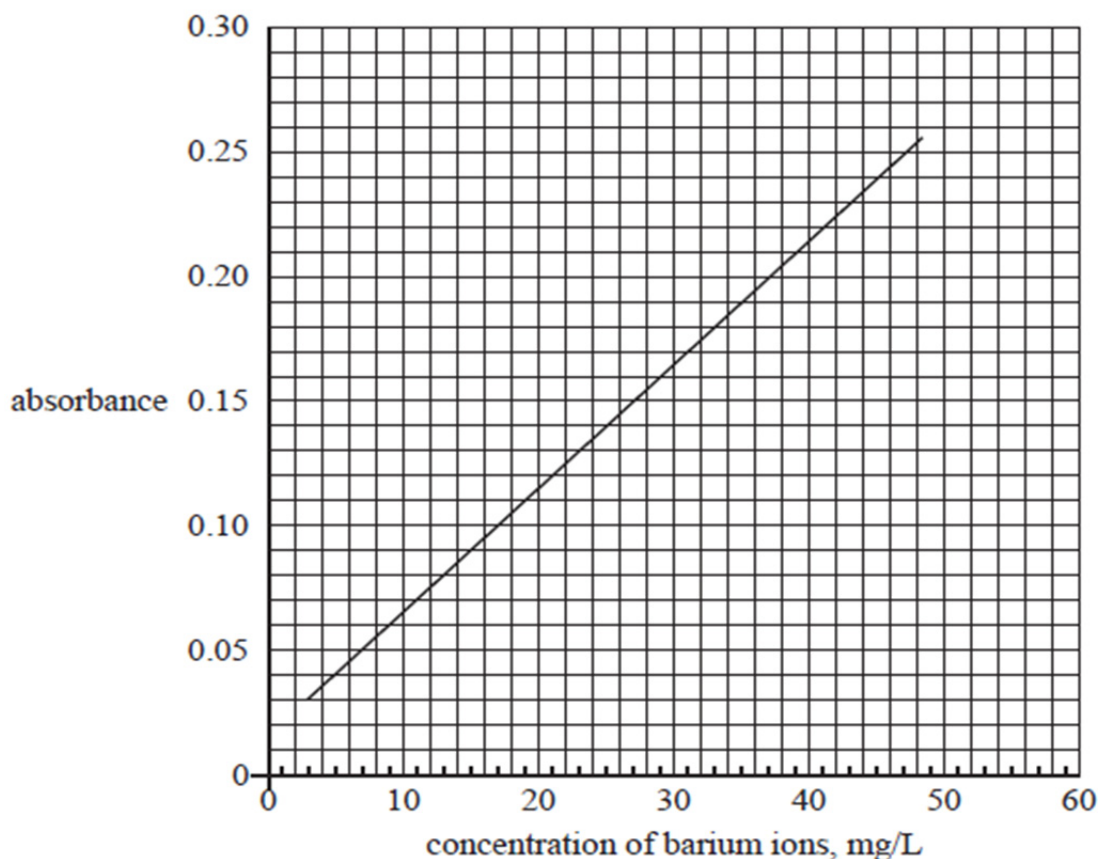
In a particular analysis, 100.0 g of grapes are treated with 100.0 mL of surfactant solution to absorb the sulfur remaining on the grapes when they were harvested.

25.00 mL of this surfactant solution is then treated to convert all of the sulfur to sulfate ions and then dried to produce a powder containing the sulfate ions.

This powder is transferred to a flask containing 2.00 mL of 200 mgL⁻¹ solution of barium Ba²⁺ ions.

The volume of solution in the flask is then made up to 10mL. A precipitate of BaSO₄ forms and settles to the bottom of the flask.

A calibration curve was prepared using standard solutions of 10, 20, 30 and 40 mgL⁻¹ Ba²⁺_(aq).



See next page

- (g) A small amount of the solution containing the unreacted Ba^{2+} ions is removed from the flask and analysed using atomic absorption spectroscopy (AAS). This solution gave an absorbance reading of 0.11.

Use the graph above to determine the concentration of barium ions remaining per 1 L of solution. Hence determine the mass of barium ions, in mg, remaining in **the 10.00 mL sample** solution.

(2 marks)

- (h) Determine the amount of barium ions, in moles, in the original 2.00 mL of 200 mgL^{-1} solution of Ba^{2+} provided and use this value to calculate the amount of barium, in moles, that **reacted** to produce the barium sulfate precipitate. (3 marks)

- (i) Determine the mass of sulfur, in mg, remaining on the 100 g of harvested grapes. (4 marks)

One common home remedy for the treatment of mildews involves the use of 'bicarbonate of soda' (potassium or sodium hydrogencarbonate). It is believed that the spraying of this solution makes plant leaves alkaline and thus prevents the germination of mildews' spores.

(j) Write the chemical formula for the **conjugate base** of hydrogencarbonate ion. _____(1 mark)

(k) In a field study, solutions containing 5000 – 10000 ppm of NaHCO_3 were found to be the most effective in controlling mildews growth in but solutions containing over 20000 ppm NaHCO_3 were found to damage and/or severely affect leaf growth in ornamental plants.

Calculate the mass of NaHCO_3 (in grams) required to make 100.0 mL of NaHCO_3 solution with a concentration of 10000 ppm assuming that NaHCO_3 dissolves completely in water and the density of water is 1.0 g mL^{-1} . (2 marks)

END OF EXAM



See next page

Spare answer page

Question number: _____

See next page

Spare answer page

Question number: _____

See next page

Spare answer page

Question number: _____

See next page